Dynamical systems and ergodic theory

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[1]Based on the lectures by professor Marco Lenci

Contents

1 Examples of simple dynamical systems **3**
- 1.1 Important example: autonomous ODEs . . . 3
- 1.2 Non-autonomous ODEs . . . 8
- 1.3 Fixed and periodic points . . . 10
  - 1.3.1 Graphical method for the orbit of a 1D map and finding fixed points . . . 11
  - 1.3.2 Classification of fixed points . . . 12
  - 1.3.3 Examples of 1d dynamical systems . . . 14
  - 1.3.4 Characterization of fixed points for 1D maps . . . 17
  - 1.3.5 Fixed points for linear flows in more than 1D . . . 20
- 1.4 Linearized ODEs around a fixed point . . . 27
- 1.5 Linear maps . . . 31
  - 1.5.1 Exhaustive list of examples in 2D . . . 31
  - 1.5.2 Linearization of a map around a fixed point . . . 34

2 Topological dynamics **38**

3 Ergodic theory **39**

4 Possible developments **40**

Main definitions

Fundamental definition

Definition 0.1 A dynamical system (DS) over the set M(usually called phase space or reference space) is a family ${\phi^t}_{t\in\mathbb{G}}$ of transformations $\phi^t : M \to M$ with the (semi)group property:

$$\phi^0 = id_M$$
$$\phi^{t+s} = \phi^s \circ \phi^t$$

Note: $\mathbb{G}$ must be a commutative semigroup with neutral element with respect to the "+" operation. In practice, for the contents of this course, $\mathbb{G} = \mathbb{R}$, $\mathbb{R}_0^+$, $\mathbb{Z}$, $\mathbb{N}$

Observation 0.1 Differential equations not depending on time derivatives satisfy the definition.

Usually the formalism is that, say x is the state now (meaning at $t = 0$), then the state at time t is given by $\phi^t(x)$.

Definition 0.2 If $\mathbb{G} = \mathbb{R}, \mathbb{R}_0^+$, then ϕ^t is called the flow of the DS. if, instead, $\mathbb{G} = \mathbb{Z}, \mathbb{N}$ usually you rename it as $T = \phi^1$, $n = t$ where T is called a map.

Observation 0.2 This is due to the fact that by just knowing T you can have enough information to know the future of the system, since

$$\phi^t = (\phi^1)^n = \underbrace{\phi^1 \circ \phi^1 .... \circ \phi^1}_{\text{n times}} = T^n$$

Note that from now on if n is an integer exponent it means to use the transformation n times. Also, if T is invertible(bijective) we denote, for $n \in \mathbb{Z}^+$:

$$T^{-n} = \underbrace{T^{-1} \circ T^{-1} ..... \circ T^{-1}}_{\text{n times}}$$

Observe that $\forall n, m \in \mathbb{Z}$:

$$T^{n+m} = T^n \circ T^m$$

Notice that by having T be invertible, it means that by knowing it you can also go back in time, and as such you know everything about the system; we will see both bijective and not in this course. (here he drew a figure).

Chapter 1

Examples of simple dynamical systems

1.1 Important example: autonomous ODEs

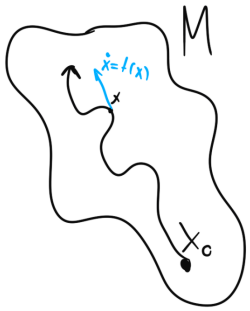


Figure 1.1: Representation of the quantities in an autonomous ODE.

The most obvious example of a dynamical system is given by an ODE, which takes the form of:

$$\dot{x} = f(x)$$

where $M \subseteq \mathbb{R}^d$, $f : M \to \mathbb{R}^d$.

We are looking at a curve of which we know the first time derivative, and we know said derivative to be a function of position, but not of time, as shown in Figure 1.1.

The aforementioned problem has, potentially, infinite solutions. What interests us are what are called Cauchy problems or, more commonly,[1] IVP, Initial

[1] This is true for the whole world, apart from Italy and France, which only call those Cauchy problems. The latter has obvious reasons to do so. Italy, on the other hand, hasn't got so many.

Value Problems.

Definition 1.1 (Cauchy problem) A Cauchy problem is given by the following linear system:

$$\begin{cases} \dot{x} = f(x) \\ x(0) = x_0 \end{cases} \tag{1.1}$$

Cauchy's theorem tells us that under some conditions, which we assume, the solution exists, and it's unique. (those conditions being that f(x) is a Lipschitz function).

We shall start with the classic ODE example: the harmonic oscillator.

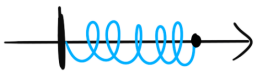


Figure 1.2: Representation of a harmonic oscillator.

Example 1.1 (Harmonic oscillator) The one dimensional harmonic oscillator is a physical system in which a mass is attracted to the origin by a force that depends on the distance to the origin itself, as shown in Figure 1.2. This is described by:

$$\ddot{x} = -w^2 x \tag{1.2}$$

At first glance, it may look like it does not fit the definition of a Cauchy problem, since it is a second derivative instead of a first one. Turns out, all high order ODEs may be reduced to a single order one.

The trick is to rename the first derivative. in this case:

$$\begin{cases} \dot{x} = v \\ \dot{v} = -w^2 x \end{cases}$$

Or, in matrix form:

$$\begin{bmatrix} \dot{x} \\ v \end{bmatrix} = \begin{bmatrix} v \\ -w^2 x \end{bmatrix} = f\left(\begin{bmatrix} x \\ v \end{bmatrix} \right)$$

We did increase the dimension of the phase space, but that does not get in the way of calling this a Cauchy problem.

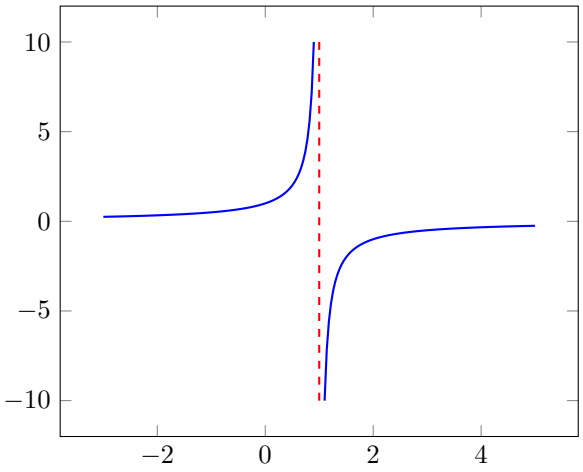
Other than uniqueness and existence, we will also be implicitly assuming all solutions to be global, i.e. defined $\forall t \in \mathbb{R}$, which is a much more restrictive condition. While finding a problem for which uniqueness and existence are not true is quite hard, the same is not true for a Cauchy problem without a global solution. Case in point:

Example 1.2 (of a non global solution)

$$\begin{cases} \dot{x} = x^2 \\ x(0) = 1 \end{cases}$$

where $M = \mathbb{R}$. Solving it by abusing math a bit:

$$\frac{dx}{dt} = x^2 \Leftrightarrow \frac{dx}{x^2} = dt \Leftrightarrow \int_1^x \frac{d\xi}{\xi^2} = \int_0^t d\tau \Leftrightarrow \left[-\frac{1}{\xi}\right]_1^x = t \Leftrightarrow 1 - \frac{1}{x} = t \Leftrightarrow x = \frac{1}{1-t}$$



As you can see, it blows up asymptotically at 1, so it is not a global solution.

We shall now see the concept of flow.

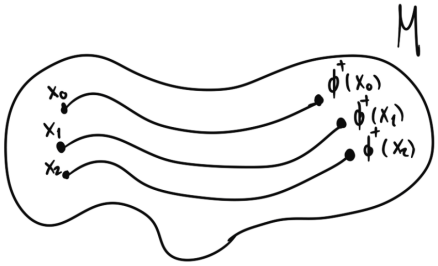


Figure 1.3: Representation of flow.

Definition 1.2 If $x(t; x_0)$ is the solution of Equation 1.1, the flow of such system is:

$$\phi^t(x_0) = x(t; x_0) \tag{1.3}$$

Proposition 1.1 *$\phi^t_{t\in\mathbb{R}}$, as defined in Equation 1.3 satisfies the group property.*

Proof. The group properties are two:

$$\phi^0 = id_M \tag{1.4}$$

$$\phi^{t+s} = \phi^s \circ \phi^t \tag{1.5}$$

Equation 1.4 is trivial, because by definition $\phi^0(x_0) = x_0 \forall\, x_0 \in M$.

The second property is a bit harder. First of all, by definition:

$$\phi^{t+s} = \phi^t \circ \phi^s \Leftrightarrow \phi^t(\phi^s(x_0)) = \phi^{t+s}(x_0) \forall x_0 \tag{1.6}$$

We define $y_1(t) = \phi^t(\phi^s(x_0))$ and $y_2(t) = \phi^{t+s}(x_0)$. We'll prove that $y_1(t)$ and $y_2(t)$ are both solutions of the following IVP:

$$\begin{cases} \dot{y}(t) = f(y(t)) \\ y(0) = \phi^s(x_0) \forall s \in \mathbb{R},\ x_0 \in M \end{cases} \tag{1.7}$$

Because of Cauchy's theorem of uniqueness, if both solve the IVP, they must be equal, thus the proposition will be proved because of Equation 1.6.

The initial condition is true by definition:

$$\begin{aligned} y_1(0) = \phi^0(\phi^s(x_0)) = \phi^s(x_0) \\ y_2(0) = \phi^{0+s}(x_0) = \phi^s(x_0) \end{aligned} \Leftrightarrow y_1(0) = y_2(0) = \phi^s(x_0) \forall s \in \mathbb{R}$$

As for the other condition, for y_1 it's true by definition, first by the definition of y_1 itself, then by the definition of $\phi^t(x_0)$:

$$\frac{dy_1}{dt}(t) = \frac{d\phi^t}{dt}(\phi^s(x_0)) = \frac{dx(t;\, \phi_s(x_0))}{dt} = \\ f(x(t;\, \phi^s(x_0))) = f(\phi^t(\phi^s(x_0))) = f(y_1(t))$$

where x is the solution of the IVP over which the flow is defined, while f() for which $\dot{x} = f(x)$ is true.

In the same way, by employing the definitions of y_2 and flow, aided by the chain rule, we may find that the temporal derivative of y_2 is equal to $f(y_2)$, where f() is the same as the one for which that is true for y_1:

$$\frac{dy_2}{dt}(t) = \frac{d\phi^{t+s}}{dt}(x_0) = \frac{dx(t+s;\, x_0)}{d(t+s)} \frac{d(t+s)}{dt} = \\ f(x(t+s;\, x_0)) = f(\phi^{t+s}(x_0)) = f(y_2(t))$$

So, y_1 and y_2 both solve the aforementioned IVP for every t, thus we proved the proposition.

□

Let's now look a bit into the concept of flow. The flow behaves in such a way(this is why it's called a flow!) in which it never intercepts. It looks, if drawn in a stylized way, as a series of lines that never intercept, because of the uniqueness of the solution of the Cauchy problem, as shown in Figure 1.3.

Proposition 1.2 *We define two flows $\phi^{t_1}(x_1)$ and $\phi^{t_2}(x_2)$. If $t_1 \geq t_2$ and $x_1 \neq x_2$, they shall never intercept.*

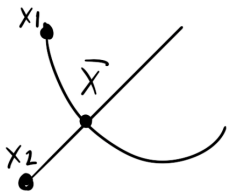


Figure 1.4: Representation of the hypothesis used to prove Proposition 1.2.

Proof. RAA[2] Assume that we have two trajectories that are completely distinct apart from one singular interception. Let's call the singular interception $\bar{x}$. The following is true by definition:

$$\bar{x} = \phi^{t_1}(x_1) = \phi^{t_2}(x_2)$$

But then, by definition of flow:

$$\phi^{-t_2}(\bar{x}) = \phi^{-t_2}(\phi^{t_2}(\bar{x})) = x_2$$

but at the same time:

$$\phi^{-t_2}(\bar{x}) = \phi^{-t_2}(\phi^{t_1}(x_1)) = \phi^{t_1 - t_2}(x_1)$$

The hypothesis here that the interception is just one. Starting from said hypothesis, we found another interception, namely x_2. This would mean, because of the uniqueness of the solution of the Cauchy problem, that the trajectories are exactly the same: but that is impossible, because we assumed that they were distinct apart from one singular point. □

Now: keep in mind that a global solution is not hard to imagine.

Example 1.3 (of global solution ascertained by visual inspection) Suppose M is a two-dimensional domain and suppose you know that at the boundary the vector field f always points inside. And suppose you know that this vector field is smooth and defined inside all the domain. Then you are assured that the solution is global. This because the vector field cannot "escape" the domain, and it covers all of it.

Usually you can find a global solution by iteratively move inside small intervals.

Now, for a little result that is more of a curiosity than anything else.

Proposition 1.3 *If ϕ^t, a continuous DS on $M \subseteq \mathbb{R}^d$, is smooth enough[3], then it is the flow of this ODE:*

$$\dot{x} = \left.\frac{\partial \phi^t}{\partial t}\right|_{t=0}$$

[2] Reductio ad absurdum. I will use this to signal that I am proving by contradiction.
[3] Meaning C^1 Lipschitz, in this case.

Proof. As it is a continuous DS, it has to satisfy the semi-group property.

Keeping this in mind, to verify the proposition we need to define the function

$$x(t,\, x_0) : t \to \phi^t(x_0)$$

for all $x_0 \in M$, then verify that it satisfies the ODE, given the initial condition:

$$x(0,\, x_0) = x_0$$

The initial condition is true by definition, because $x(0, x_0) = \phi^0(x_0) = x_0$.

Then, we must verify that it does indeed solve the ODE. By definition:

$$\dot{x}(t, x_0) = \frac{\partial \phi^t(x_0)}{\partial t}$$

then, by employing the chain rule[4]:

$$\frac{\partial \phi^t(x_0)}{\partial t} = \frac{\partial \phi^{t+s}(x_0)}{\partial s}\bigg|_{s=0} = \frac{\partial}{\partial s}\phi^s(\phi^t(x_0))\bigg|_{s=0} = \frac{\partial \phi^s}{\partial s}(\phi^t(x_0))\bigg|_{s=0} = \frac{\partial \phi^s}{\partial s}(x(t, x_0))\bigg|_{s=0}$$

Then, straightforwardly:

$$\dot{x}(t, x_0) = \frac{\partial \phi^s}{\partial s}(x(t, x_0))\bigg|_{s=0} = \frac{\partial \phi^t}{\partial t}(x(t, x_0))\bigg|_{t=0}$$

As ugly as that may be, it is true, since s is just a name: we can change s to t without second thoughts because it is a partial derivative, which fixes $x(t, x_0)$ in place, thus avoiding uncomfortable situations with the temporal dependency of the position. □

1.2 Non-autonomous ODEs

Definition 1.3 A non-autonomous ODE is a differential equation of the form:

$$\dot{x} = f(x,\, t)$$

This kind of problem brings with itself some problems, mainly that you cannot use the theory of dynamical systems. Let's see why.

[4]Not immediately obvious:

$$\frac{\partial \phi^{t+s}(x_0)}{\partial s} = \frac{\partial \phi^{t+s}(x_0)}{\partial (t+s)}\frac{\partial (t+s)}{\partial s} = \frac{\partial \phi^{t+s}(x_0)}{\partial (t+s)}$$

Now, if we evaluate that at $s = 0$:

$$\frac{\partial \phi^{t+s}(x_0)}{\partial s}\bigg|_{s=0} = \frac{\partial \phi^{t+s}(x_0)}{\partial (t+s)}\bigg|_{s=0} = \frac{\partial \phi^t(x_0)}{\partial t}$$

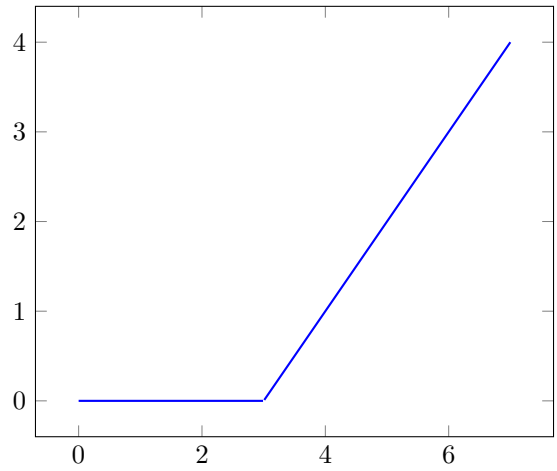


Figure 1.5: Plot of f(x, t) for the example.

Example 1.4 Let's say that:

$$f(x,\, t) = \begin{cases} 0 & 0 < x < 3 \\ x - 3 & 3 < x \end{cases}$$

Calling the flow $\phi^t(x_0) = x(t,\, x_0)$, we could easily see that the semi-group property is not satisfied. Say that you apply ϕ^2 twice:

$$\phi^2(x_0) = x_0 \Leftrightarrow \phi^2(\phi^2(x_0)) = \phi^2(x_0) = x_0$$

quite trivially:

$$\phi^4(x_0) = x_0 + 1 \Leftrightarrow \phi^4 \neq \phi^2(\phi^2(x_0))$$

This renders absolutely impossible to use DS theory, at first glance. How do you go around it?

The first way one could think of is to notice that the IVP's initial condition is really not that much of a constraint. Let's see why this is not true.

Proposition 1.4 *We define an IVP as follows:*

$$\begin{cases} \dot{x} = f(x) \\ x(t_0 \neq 0) = x_0 \end{cases}$$

we define $\phi^{t_0,t}(x_0) = x(t, x_0, t_0)$. *In this case, you lose the semi-group property, i.e.* $\phi^{t,s} \circ \phi^{t_0,t} = \phi^{t_0,s}$.

Proof. Homework. □

There is a correct workaround, and it is the following.

Proposition 1.5 *Suppose you have the following IVP, where $x \in M \subseteq \mathbb{R}^d$:*

$$\begin{cases} \dot{x} = f(x,t) \\ x(0) = x_0 \end{cases}$$

Which is built on a non-autonomous ODE. We set $y = (x,t) \in M \times \mathbb{R} \subseteq \mathbb{R}^{d+1}$ and $F(y) = F(x,t) = (f(x,t),1)$.

Then, we consider the IVP:

$$\begin{cases} \dot{y} = F(y) \\ y(0) = (x_0, 0) \end{cases}$$

Where the dot is the derivative with respect to the newly defined "time" s, for which the following is true:

$$\begin{bmatrix} \frac{dx}{ds}(s) \\ \frac{dt}{ds}(s) \end{bmatrix} = \begin{bmatrix} f(x(s),t(s)) \\ 1 \end{bmatrix}$$
$$\begin{bmatrix} x(0) \\ t(0) \end{bmatrix} = \begin{bmatrix} x_0 \\ 0 \end{bmatrix}$$

Then this is an autonomous ODE, with respect to t.

Proof. First things first, we can actually see how s behaves in respect to t from the bottom members of the matrices of its definition:

$$\frac{dt}{ds}(s) = 1, t(0) = 0 \Leftrightarrow t(s) = s$$

so s and t are actually the same. So, by looking at the top members, you find the previous problem with respect to t. □

Observation 1.1 What we are doing is the classic physics trick to consider time as a dimension as the other ones.

So, basically, if you have a non-autonomous ODE you can make it an autonomous one.

1.3 Fixed and periodic points

Definition 1.4 Considering the usual $\mathbb{G} = \mathbb{R}, \mathbb{R}_0^+, \mathbb{Z}, \mathbb{N}$, for a DS ${\phi^t}_{t \in \mathbb{G}}$, the orbit or trajectory of $x \in M$ is defined as:

$$O(x) = \{\phi^t(x) | t \in \mathbb{G}\}$$

In the cases of interest, we will use

$$O^{t(-)}(x) = \{\phi^t(x) | t \geq (\leq) 0\}$$

and call it future or forward (past or backward, in the case with minus and $\leq$) orbit or trajectory.

Definition 1.5 $\bar{x} \in M$ is called a fixed point if $\phi^t(\bar{x}) = \bar{x}, \forall t > 0$.

Else, $\bar{x}$ is called a periodic point if $\exists T > 0 | \phi^T(\bar{x}) = \bar{x}$, and T is called a period for $\bar{x}$.

In the case of maps, the set of fixed points is sometimes denoted $Fix(T) = x \in M | T(x) = x$.

Observation 1.2 Clearly, in periodic cases $\phi^{t+T}(\bar{x}) = \phi^t(\bar{x})$.[5]

Observation 1.3 Notice that if $x \in Fix(T)$, then $T^n(x) = x$ which is true if and only if $x \in Fix(T^n)$; more formally:

$$x \in Fix(T) \Rightarrow T^n(x) = x \Leftrightarrow x \in Fix(T^n)$$

Also, one can very easily notice that a periodic point of T is by definition a point in $Fix(T^n)$ for some $n \in \mathbb{Z}$.

1.3.1 Graphical method for the orbit of a 1D map and finding fixed points

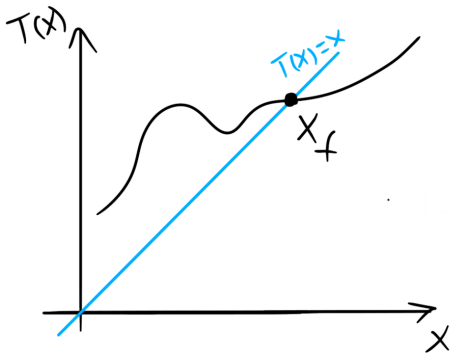


Figure 1.6: Plot of an arbitrary invertible map and its intersection with the bisectrix of the first and third plane, where x_f is a fixed point.

Example 1.5 (1D maps) Consider the following kind of map:

$$T : M \subseteq \mathbb{R} \to M$$

We want to find the fixed points. By definition, what you are trying to find is the points for which $x = T(x)$. Graphically, this means that by plotting $T(x)$ and then the bisectrix of the first and the third plane, and then finding their interceptions, you obtain the fixed points, as shown in Figure 1.6.

Riffing from this simple method, one can develop a really easy graphical method to find the orbit of a 1D map, which works as follows, as shown in Figure 1.7:

[5] Provable by group property on the definition.

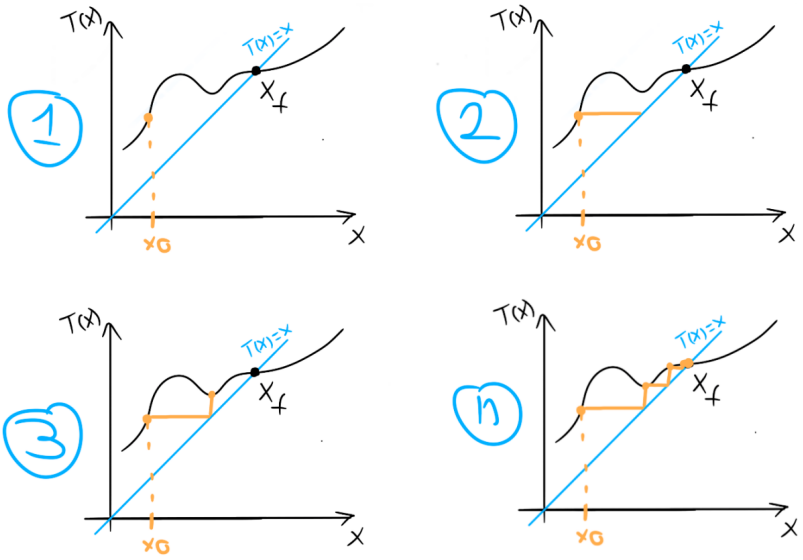


Figure 1.7: Representation of the algorithm after the first steps and when fully executed, on an arbitrary map.

1. Find $T(x_0)$ on the graph;
2. Draw a horizontal line from $T(x_0)$ to the bisectrix of the first and third plane;
3. From the intersection of the bisectrix and the just drawn horizontal line, draw a vertical one until intersecting T(x);
4. Repeat.

Observation 1.4 In the represented case, we find that the orbit converges to the fixed point, but this is not always true, as shown in Figure 1.8. Just from the two examples we did, we may already infer that how the orbit behaves may have something to do with the derivative of the map. We will look further into that during this course.

1.3.2 Classification of fixed points

Here, we will assume that M is a topological space.

Definition 1.6 A topological space is one in which the notions of open sets, closed sets and neighbourhood are defined.[6]

Definition 1.7 A fixed point $\bar{x} \in M$ is called:

[6] Iffy as hell as a definition.

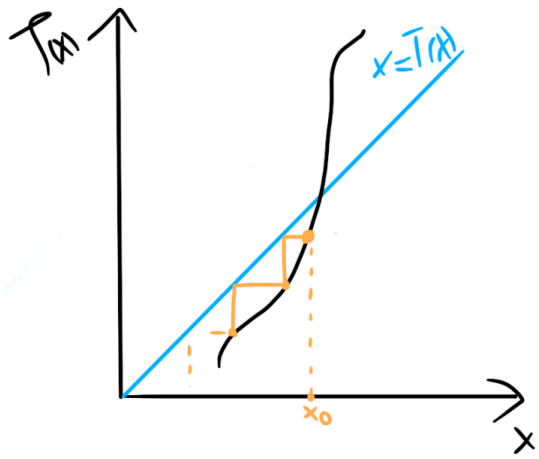


Figure 1.8: Representation of the algorithm after some steps executed on another map.

- **stable** if for every neighbourhood U of $\bar{x}$ exists a neighbourhood V[7] of $\bar{x}$ such that for every $x \in V$, $\phi^t(x)$ is in U for every positive t[8]. In math:

$$\forall U \text{ nbhd of } \bar{x}\, \exists V \text{ nbhd of } \bar{x} | \forall x \in V, \phi^t(x) \in U, \forall x \geq 0$$

- **attractive** (or attracting) if it is stable and for every $x \in V$, $\phi^t(x)$ tends to $\bar{x}$ for infinite time. In math:

$$\forall x \in V, \lim_{t \to + \inf} \phi^t(x) = \bar{x}$$

- **unstable** if it is not stable, i.e.[9] there is a neighbourhood U of $\bar{x}$ such that for every neighbourhood V of $\bar{x}$, exists and $x \in V$ and a $\bar{t}$ which is a function of said x $\bar{t}(x)$ such that $\phi^{\bar{t}}(x)$ is not in U. In math:

$$\exists U \text{ nbhd of } \bar{x} | \forall V \text{ nbhd of } \bar{x}, \exists x \in V, \bar{t} = \bar{t}(x) | \phi^{\bar{t}}(x) \notin U$$

- **repulsive** (or repelling) if there is a U neighbourhood of $\bar{x}$ such that for every $x \in U$ there is a $\bar{t}$ which is a function of x $\bar{t}(x)$ such that $\phi^{\bar{t}}(x)$ is not in U. In math:

$$\exists U \text{ nbhd of } \bar{x} | \forall x \in U \exists \bar{t} = \bar{t}(x) | \phi^{\bar{t}}(x) \notin U$$

All of these definitions are exemplified by the drawings in Figure 1.9.

[7] can be trivially proven to be inside U

[8] This means that if you look close enough, you can look even closer and say if I start there I will stay close in the future.

[9] Id est, which is Latin for "that is".

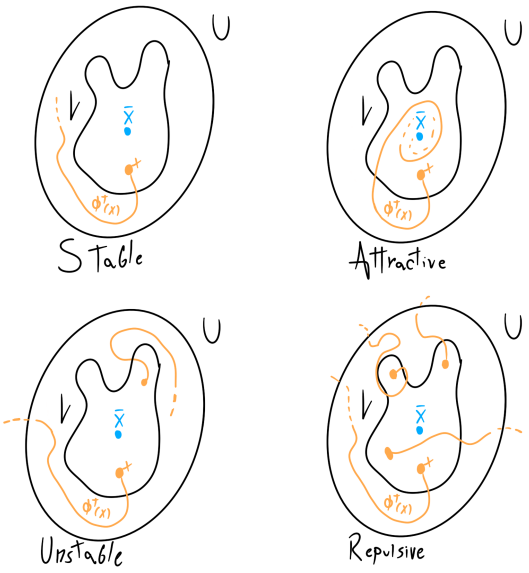


Figure 1.9: Representation of the types of fixed points.

1.3.3 Examples of 1d dynamical systems

Let's now try to tackle some actual dynamical systems; the following examples are proposed models of population growth.

Example 1.6 (by Thomas Malthus) In 1798, Thomas Malthus proposed the following model while trying to model the growth of population:

$$\dot{x}(t) = \lambda x(t) \Rightarrow x(t) = x_0 e^{\lambda t} (t \in \mathbb{R}, \lambda \in \mathbb{R})$$

Alternatively, with discrete time:

$$x_{n-1} = \alpha x_n (\alpha \in \mathbb{R}^+)$$

which, using the introduced terminology, means that the map is defined as follows:

$$\begin{aligned} T \colon \mathbb{R}^+ &\longrightarrow \mathbb{R}^+ \\ x &\mapsto \alpha x, \end{aligned}$$

This is not hard to solve, and the solution is:

$$x_n = x_0 \alpha^n$$

Heuristically, this means that the growth of the population is proportional to the number of people, which is reasonable.

One quick note: oftentimes discrete time is more realistic than real time, since you cannot actually measure the population at all times.

Those models are rarely realistic, though. There are more advanced ones.

Example 1.7 (by Pierre Francois Velhurst) Popularized by Robert May in 1978, the logistic model was developed by P.F. Velhurst in 1838 as means of modeling population growth.

$$x_{n+1} = \alpha(A - x_n)x_n$$

The new modeling "step" is adding a sort of cap to the amount of people an ecosystem can support. Once that cap, which is parametrized in the equation by A, is surpassed, the population will have trouble surviving. This can be understood considering $\alpha(A - x_n)$ as growth coefficient: it changes in respect with the distance of the real population with the cap, also factually limiting the population: it would not make sense for $\alpha(A - x_n)$ to be negative since (differently from the continuous case) having that coefficient to be negative would make x_{n+1} negative, which does not make sense.

We want to know the map of this system. We shall start by rearranging the terms and then changing variable:

$$\frac{x_{n+1}}{A} = \alpha\left(1 - \frac{x_n}{A}\right)\frac{x_n}{A}$$

Let's call $y = \frac{x}{A}$. Then:

$$y_{n+1} = ry_n(1 - y_n), \qquad r = A\alpha$$

We may now define:

$$\begin{aligned} T\colon [0,1] &\longrightarrow [0,1] \\ x &\mapsto \alpha r x(1-x), \end{aligned}$$

The justification for the domain to be in $[0, 1]$ is obviously that, as stated before, $0 \leq x \leq A$ so $0 \leq \frac{x}{A} \leq 1$.

As we may see in Figure 1.10, the maximum occurs when $x = \frac{1}{2}$, which means that the maximum is $T(\frac{1}{2}) = \frac{r}{4}$. Since we want it to be a self-map(domain and codomain are the same space), we have the added constraint $r \in [0, 4]$.

We are now interested in finding the fixed points of this map. Luckily, this is one of the rare cases in which we may find the exact solutions, which we will call x_t and x_f[10]:

$$T(x) = x \Leftrightarrow -rx^2 + rx = x \Leftrightarrow -rx\left(x - \frac{r-1}{r}\right) = 0 \Rightarrow \begin{cases} x_t = 0 \\ x_f = \frac{r-1}{r} \end{cases}$$

Note that x_f only "exists" for $r \geq 1$, since we are in [0,1].

[10]For trivial and fixed.

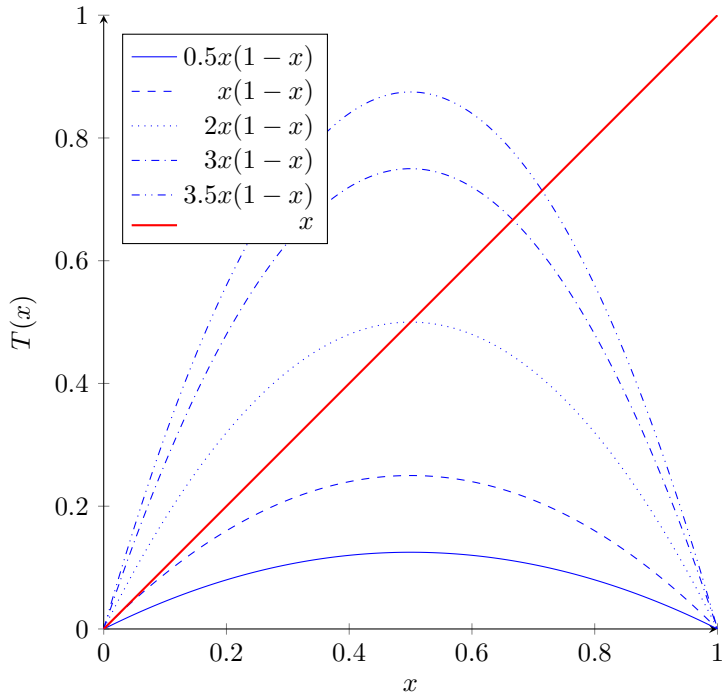


Figure 1.10: Plot of the logistic map and the bisectrix with varying r.

We had a hunch, when we found the graphical method for the orbit, that the nature of the fixed point was related in some way with the value of the derivative of the map, more precisely with it being more or less than 1. We will prove this in general in the future, but for the time being, let's just calculate the derivative of this map:

$$T'(x) = -2rx + r \Rightarrow \begin{cases} T'(x_t) = r \\ T'(x_f) = -2r\frac{r-1}{r} + r = 2 - r \end{cases}$$

Which, in turn, implies that:

$$|T'(0)| < 1 \Leftrightarrow r < 1 \text{ and } |T'(x_f)| < 1 \Leftrightarrow 1 < r < 3$$

So, we may expect different behaviors in those four different regions of the domain of $r \in [0, 4]$, namely $[0, 1]$, $[1, 3]$, $[3, 4]$. Let's see how true that is.

(algoritmo grafico sui cosi per i vari casi)

- $r \leq 1$: when r is less than 1, the only fixed point is given by $x_t = 0$. In this case, the orbit converges quite quickly towards the origin;
- $1 < r \leq 2$: when r is between 1 and 2, the maximum is below the bisectrix (apart from $r = 2$), which implies that x_f is in the first half of the map,

or $x_f \leq 0.5$. In this case, all the trajectories converge to x_f, apart from the one that already starts at $x_t = 0$;

- $2 < r \leq 3$: when r increases (without surpassing 3) the behaviour is the same: the orbit converges towards x_f if $x_0 \neq 0$ but in a different way, namely it oscillates around x_f until it reaches it, overshooting and undershooting it until finally converging;

- $3 < r \lessapprox 3.45$: increasing further, we can see that the orbit becomes periodic, oscillating between two points with $T = 2$. So, it may be interesting to see the behaviour of the squared map, where we can see that there are 4 periodic points (namely x_t, x_f and the two points between which it oscillates). We may also notice that the trajectory there tends to one of the two periodic points (if you don't start either in x_t or x_f), which is a matter of "scope": the squared map only sees one of the two points in the periodic motion, thus those are attractive periodic points for the orbit as seen by $T^2(x)$;

- $r \gtrapprox 3.45$: once you go over circa 3.45, the movement becomes much more unpredictable: it is still periodic, but the periodicity will increase a lot even with small variations of r, and as such the motion will becomes much more chaotic;

All of those situations are shown below.

If one plots the attractive fixed points (for some n in T^n), as shown in Figure 1.12 what will be observed is that, while for $0 < r < 1$ the only attractive point is 0, and for $1 < r < 3$ it varies but is still a single point, then the number starts increasing by bifurcations (this phenomenon is called period doubling). It is not in the scope of this course to define and analyze this pattern and see if it is, or it is not chaotic. Just know that we are seeing a pattern highly determined by the initial conditions.

1.3.4 Characterization of fixed points for 1D maps

We shall now demonstrate the hunch we had about the correlation between the derivative and the nature of the fixed point, at least in 1D.

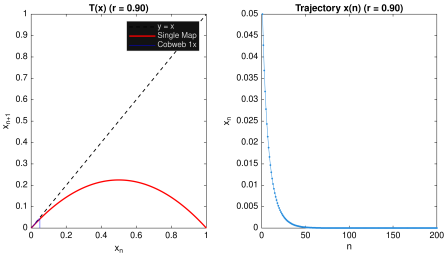
Proposition 1.6 *Let $T : I \to I$, I connected set[11] in $\mathbb{R}$, $T \in C^1(I)$. Let $\bar{x} \in Fix(T)$. Then:*

1. *if $|T'(\bar{x})| < 1$ then $\bar{x}$ is attractive;*

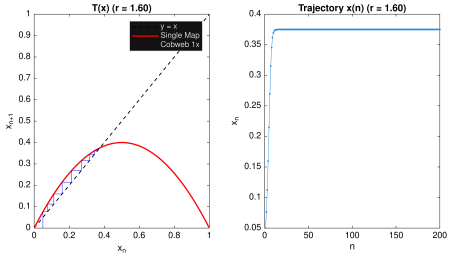
2. *if $|T'(\bar{x})| > 1$ then $\bar{x}$ is repelling;*

[disegni del diagramma di cobweb]

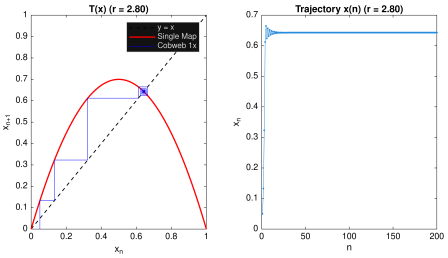
[11] Meaning either an interval, half line, or $\mathbb{R}$.



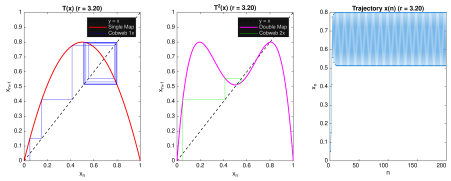
(a) $r = 0.90$



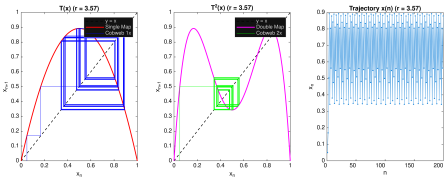
(b) $r = 1.60$



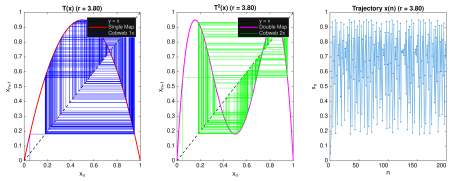
(c) $r = 2.80$



(d) $r = 3.20$



(e) $r = 3.57$



(f) $r = 3.80$

Figure 1.11: Maps and trajectories with varying r.

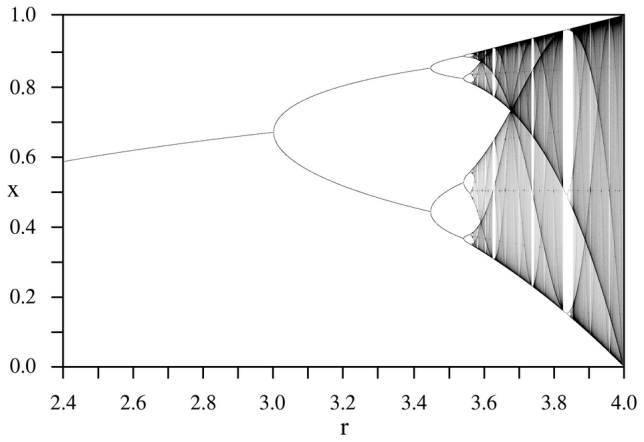


Figure 1.12: Bifurcation graph for the logistic map.

1. Since $T \in C^1$, the derivative is continuous, so there exists a neighbourhood $U = (\bar{x} - \delta, \bar{x} + \delta)$[12] such that for every $x \in U, |T'(x)| \leq \lambda < 1$. In math:

$$T \in C^1, T'(\bar{x}) < 1 \Rightarrow \exists U \textbf{ nbhd of } \bar{x} | \forall x \in U, |T'(x)| \leq \lambda < 1, \lambda \text{ is close to 1.}$$

Let $x_0 \in U$ be an arbitrary point in the neighbourhood, and we shall call $x_n = T^n(x_0)$.

We can notice that, if for all $n \geq 1$, x_n is in U and $|x_n - \bar{x}| \leq \lambda^n |x_0 - \bar{x}|$, or:

$$n \geq 1 \Rightarrow x_n \in U \wedge |x_n - \bar{x}| \leq \lambda^n |x_0 - \bar{x}| \tag{1.8}$$

then $x_n \to x_0$, which would prove that the point is attractive. Thus, to prove the proposition we need to prove Equation 1.8.

This can be done by induction. First, let' say that $n = 1$. In that case, by definition:

$$|x_1 - \bar{x}| = |T(x_0) - T(\bar{x})|$$

by using Lagrange's mean value theorem[13] we can find that, for some $\psi \in (x_0, \bar{x}) \in U$

$$|T(x_0) - T(\bar{x})| = |T'(\phi)(x_0 - \bar{x})| \leq \lambda |x_0 - \bar{x}|$$

which is a part of the claim. Also, by cancelling the passages:

$$|x_1 - \bar{x}| \leq \lambda |x_0 - \bar{x}| \Rightarrow x_1 \in U$$

[12] $]\bar{x} - \delta, \bar{x} + \delta[$

[13] Let $f \colon [a, b] \to \mathbb{R}$ be $f \in C^1$. Then $\exists\, \psi \in (a, b)$ such that:

$$f'(\psi) = \frac{f(b) - f(a)}{b - a}$$

which completes the verification for n=1.

We are left to prove that if the statement is true for n, then it is true for n+1. The process up to a point is exactly the same:

$$|x_{n+1} - \bar{x}| = |T(x_n) - T(\bar{x})| = |T'(\psi_n)(x_n - \bar{x})| \leq \lambda |x_n - \bar{x}|$$

Now, we can find by iteration that $\lambda |x_n - \bar{x}| \leq \lambda^{n+1} |x_0 - \bar{x}|$, which what we were trying to demonstrate. $x_{n+1} \in U$ is trivially proved in the same way as the first part. □

2. Homework. □

Definition 1.8 If $|T'(\bar{x})| = 1$ the fixed point is called neutral or indifferent.

1.3.5 Fixed points for linear flows in more than 1D

Say you have a linear ODE with constant data:

$$\begin{cases} \dot{x} = Ax, \qquad x \in \mathbb{R}^d, \text{ A is a constant matrix} \in \mathbb{R}^{d \times d} \\ x(0) = x_0 \in \mathbb{R}^d \end{cases} \tag{1.9}$$

The solution of the initial value problem is:

$$x(t) = e^{tA} x_0, \, \forall A \in \mathbb{R}^{d \times d}$$

Where the exponential is defined as follows.

Definition 1.9 If B is a matrix, then $e^B = \sum_{k=0}^{\infty} \frac{B^k}{k!}$. Note that the series converges in the space of $d \times d$ matrices with respect to any norm, because if $||B|| < \infty \Rightarrow ||B^k|| \leq ||B||^k$. [14]

Observation 1.5 There is no guarantee that A is diagonalizable. One could point out that you can prove that finding non-diagonalizable matrices is quite hard, and even then they are infinitely "close" to the diagonalizable ones, but we do not care about the space of matrices per se, we care about single matrices individually. So we have to consider the possibility that our matrix is not diagonalizable.

Luckily, linear algebra comes to our aid, thanks to Jordan[15]'s theorem.

[14] The definition might sound a bit iffy, in the sense that the whole "in any norm" thing is a big leap of faith. Actually, in finite dimensions, if $|\,|, ||\,||, |||\,|||$ are all norms of V, $dim(V) < \infty$, then there exists a $C > 1$ such that

$$C^{-1}|v| \leq ||v|| \leq C||v||$$

so, if you prove a limit with some norm it is actually true for all of them due to the definition of norm.

[15] Six FMVPs, six scoring titles, five MVPs, one DPOY, 10 times All-NBA, nine All-defensive team, three ASGMVP. Greatest PPG in the history of the NBA, both in the regular season and in the playoffs, and an important theorem to his name. Lebron ain't got no theorem to his name.

Theorem 1.1 (Jordan's theorem) *Any matrix $A \in \mathbb{R}^{d\times d}$ can be put in Jordan form:*

$$\begin{bmatrix} J_1 & 0 & \cdots & 0 \\ 0 & J_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & J_k \end{bmatrix}$$

where a Jordan block[16] *J_i takes the form:*

$$J_i = \begin{bmatrix} \lambda_i & 1 & 0 & \cdots & 0 \\ 0 & \lambda_i & 1 & \cdots & 0 \\ 0 & 0 & \lambda_i & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & 1 \\ 0 & 0 & 0 & \cdots & \lambda_i \end{bmatrix}$$

Note that Jordan blocks can be of different sizes.

Also, there exists a matrix $C \in \mathbb{C}^{d\times d}$ with a non-zero determinant, such that $A = CA_JC^{-1}$. C is called the base change matrix.

For now on, we will always assume A is already in Jordan form. This is rapidly justified.

Proposition 1.7 *Starting from the problem given by Equation 1.9, one may find an equivalent problem with the Jordan form of A.*

Proof. Starting from Jordan's theorem:

$$A = CA_JC^{-1} \Leftrightarrow A_J = C^{-1}AC$$

which implies:

$$\dot{x} = Ax \Leftrightarrow \dot{x} = CA_JC^{-1}x \Leftrightarrow C^{-1}\dot{x} = A_JC^{-1}x$$

So, by simply renaming $y = C^{-1}x$ and so $y_0 = C^{-1}x_0$ we may easily find:

$$\begin{cases} y = A_Jy \\ y(0) = y_0 \end{cases}$$

which is an equivalent problem. □

Examples in 2D

Example 1.8 Let the A in Equation 1.9 be:

$$A = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}, \qquad \lambda \neq 0$$

[16] https://youtu.be/tRhYR5DB2rg?si=qMxcfBx-5ecyhcjr

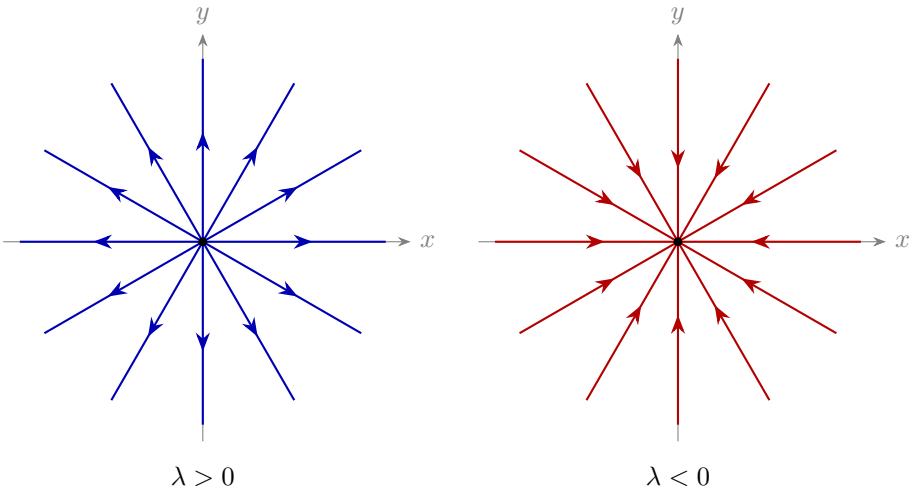
Following what we now know, the solution has the form:

$$e^{tA} = \begin{bmatrix} e^{\lambda t} & 0 \\ 0 & e^{\lambda t} \end{bmatrix}$$

Consequently (note that here x and y are the Cartesian coordinates):

$$\begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = \begin{bmatrix} x_0 e^{\lambda t} \\ y_0 e^{\lambda t} \end{bmatrix}$$

In this kind of situations, one wants to plot y as a function of x and then show the dependency on time as an arrow. In this case, as shown in Figure 1.13, the trajectories are attracted by the origin if $\lambda < 0$, else they escape from it.



$\lambda > 0$ $\lambda < 0$

Figure 1.13: Phase portrait for $A = \lambda I$. The trajectories are escaping from the origin ($\lambda > 0$) or attracted by the origin ($\lambda < 0$).

Let's spice things up a bit.

Example 1.9 Let the A in Equation 1.9 be:

$$A = \begin{bmatrix} \lambda & 0 \\ 0 & \mu \end{bmatrix}$$

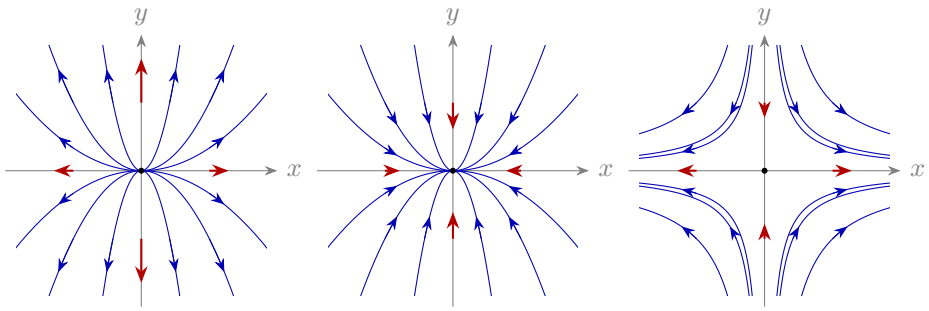
This implies:

$$\begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = \begin{bmatrix} x_0 e^{\lambda t} \\ y_0 e^{\mu t} \end{bmatrix}$$

We can rework it in order to plot it, assuming we are in the non-trivial case:

$$\lambda, \mu \neq 0, \begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = \begin{bmatrix} x_0 e^{\lambda t} \\ y_0 e^{\mu t} \end{bmatrix} \Leftrightarrow t = \frac{1}{\lambda} \ln\left(\frac{x}{x_0}\right) \Leftrightarrow y = y_0 e^{\frac{\mu}{\lambda} \ln\left(\frac{x}{x_0}\right)} = y_0 (\frac{x}{x_0})^{\frac{\mu}{\lambda}}.$$

The results of this are shown in Figure 1.14.



(a) Unstable node, $0 < \lambda < \mu$.

(b) Stable node, $\mu < \lambda < 0$.

(c) Saddle, $\lambda > 0 > \mu$.

(d) Unstable node, $0 < \mu < \lambda$.

(e) Stable node, $\lambda < \mu < 0$.

(f) Saddle, $\mu > 0 > \lambda$.

Figure 1.14: Phase portraits of $\dot{x} = \lambda x$, $\dot{y} = \mu y$. Blue curves are $y = y_0 \, (x/x_0)^{\mu/\lambda}$; red half-lines are the trajectories on the coordinate axes. Arrows indicate the direction of increasing time.

What about complex coefficients?

Example 1.10 Let the A in Equation 1.9 be:

$$A = \begin{bmatrix} a + ib & 0 \\ 0 & a - ib \end{bmatrix}, \qquad a, b \in \mathbb{R}$$

This implies, based on what was stated at the beginning of the section:

$$e^{tA} = \begin{bmatrix} e^{at}e^{ibt} & 0 \\ 0 & e^{at}e^{-ibt} \end{bmatrix}$$

since the initial condition is real, we may infer that the solution will be real. We will nonetheless prove it.

Let's do another change of variable, in order to have a real matrix.

Consider the following matrix:

$$D = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & i \\ i & 1 \end{bmatrix} \Leftrightarrow D^{-1} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -i \\ -i & 1 \end{bmatrix}$$

We shall use $z = D^{-1}y$. We can get to do so by using the fact that $DD^{-1} = D^{-1}D = I$ on $\dot{y} = Ay$ as following:

$$\dot{y} = D^{-1}DADD^{-1}y \Leftrightarrow D^{-1}\dot{y} = D^{-1}ADD^{-1}y \Leftrightarrow \dot{z} = Bz$$

where $B = D^{-1}AD$.

We are trying to find a real equation, so we need to prove that $B \in \mathbb{R}^{2\times 2}$ and $e^{tB} \in \mathbb{R}^{2\times 2}$.

First of all, $e^{tB} \in \mathbb{R}^{2\times 2}$ implies $B \in \mathbb{R}^{2\times 2}$ because:

$$\frac{d}{dt} e^{tB}\Big|_{t=0} = B$$

[17]

So we only need to prove $e^{tB} \in \mathbb{R}^{2\times 2}$. We shall start by observing that:

$$e^{tB} = e^{tD^{-1}AD} = \sum_k \frac{t^k}{k!}(D^{-1}AD)^k = D^{-1}e^{tA}D$$

because $(D^{-1}AD)^k = D^{-1}A^kD$ since the Ds simplify themselves[18]. We can then insert the actual values of D, D^{-1} and e^{tA} in that, and then compute the value:

$$D^{-1}e^{tA}D = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -i \\ -i & 1 \end{bmatrix} \begin{bmatrix} e^{at}e^{ibt} & 0 \\ 0 & e^{at}e^{-ibt} \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & i \\ i & 1 \end{bmatrix} =$$
$$\frac{e^{at}}{2} \begin{bmatrix} e^{ibt} & -ie^{-ibt} \\ -ie^{ibt} & e^{-ibt} \end{bmatrix} \begin{bmatrix} 1 & i \\ i & 1 \end{bmatrix} = \frac{e^{at}}{2} \begin{bmatrix} e^{ibt} + e^{-ibt} & ie^{ibt} - ie^{-ibt} \\ -ie^{ibt} + ie^{-ibt} & e^{ibt} + e^{-ibt} \end{bmatrix}$$

[17] This can be proved, if you are not convinced:

$$\frac{de^{tA}}{dt} = \frac{d}{dt}\sum_{k=0}^{\infty} \frac{t^kA^k}{k!} = \sum_{k=0}^{\infty} \frac{d}{dt}\frac{t^kA^k}{k!}$$

But $\frac{d}{dt}\frac{t^0A^0}{0!} = 0$, so we may exclude the term with $k = 0$, obtaining the following:

$$\sum_{k=0}^{\infty} \frac{d}{dt}\frac{t^kA^k}{k!} = \sum_{k=1}^{\infty} \frac{d}{dt}\frac{t^kA^k}{k!} = \sum_{k=1}^{\infty} \frac{t^{k-1}A^{k-1}}{(k-1)!}A = A\sum_{j=0}^{\infty} \frac{t^jA^j}{(j)!} = Ae^{tA}$$

which also converges with $||\,||$ in $\mathbb{R}^{d\times d}$.

It is actually as a consequence of this that $r(t) = e^{tA}r_0$ is the solution of the following IVP:

$$\begin{cases} \dot{r} = Ar \\ r(0) = r_0 \end{cases}$$

In fact, $r(0) = Ir_0 = r_0$ and

$$\dot{r}(t) = Ae^{tA}r_0 = Ar(t)$$

[18] Imagine $k = 2$. Then you have $D^{-1}ADD^{-1}AD = D^{-1}AAD$.

Which, remembering the polar form for complex numbers:

$$\frac{e^{at}}{2}\begin{bmatrix} e^{ibt}+e^{-ibt} & ie^{ibt}-ie^{-ibt} \\ -ie^{ibt}+ie^{-ibt} & e^{ibt}+e^{-ibt} \end{bmatrix} = e^{at}\begin{bmatrix} cos(bt) & -sin(bt) \\ sin(bt) & cos(bt) \end{bmatrix}$$

So we have proved that

$$D^{-1}e^{tA}D = e^{at}\begin{bmatrix} cos(bt) & -sin(bt) \\ sin(bt) & cos(bt) \end{bmatrix}$$

which is not only a real matrix, which is what we wanted to prove, but is also the propagator, thus we have solved the IVP. The solution is, indeed, $z(t) = e^{tB}z_0$. Alternatively:

$$\begin{bmatrix} x \\ y \end{bmatrix} = e^{at}R_{bt}\begin{bmatrix} x_0 \\ y_0 \end{bmatrix}$$

where R_{bt} is a rotation matrix indicating a rotation of magnitude bt, which is the one we found. This makes it especially easy to visualize the drawings, shown in Figure 1.15.

We are missing a case: the Jordan form[19].

Example 1.11 Let the A in Equation 1.9 be:

$$A = \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix}$$

where $\lambda \in \mathbb{R}$.[20]

It can be proved that

$$A^k = \begin{bmatrix} \lambda^k & k\lambda^{k-1} \\ 0 & \lambda^k \end{bmatrix}$$

which implies:

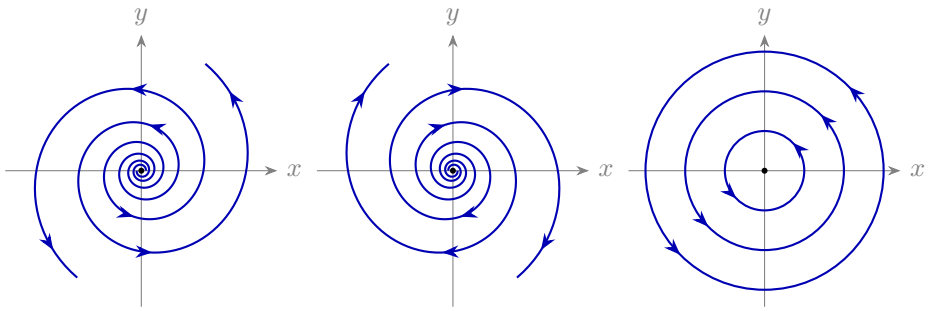
$$e^{tA} = \begin{bmatrix} \sum_{k=0}^{\infty}\frac{t^k\lambda^k}{k!} & \sum_{k=0}^{\infty}\frac{t^k k\lambda^{k-1}}{k!} \\ 0 & \sum_{k=0}^{\infty}\frac{t^k\lambda^k}{k!} \end{bmatrix} = \sum_{k=0}^{\infty}\frac{A^k}{k!} = \begin{bmatrix} \sum_{k=0}^{\infty}\frac{t^k\lambda^k}{k!} & \sum_{k=0}^{\infty}\frac{t^k\lambda^{k-1}}{(k-1)!} \\ 0 & \sum_{k=0}^{\infty}\frac{t^k\lambda^k}{k!} \end{bmatrix} = \begin{bmatrix} e^{\lambda t} & te^{\lambda t} \\ 0 & e^{\lambda t} \end{bmatrix}$$

So the general solution of

$$\begin{bmatrix} \dot{x} \\ y \end{bmatrix} = A\begin{bmatrix} x \\ y \end{bmatrix}, \begin{bmatrix} x(0) \\ y(0) \end{bmatrix} = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}$$

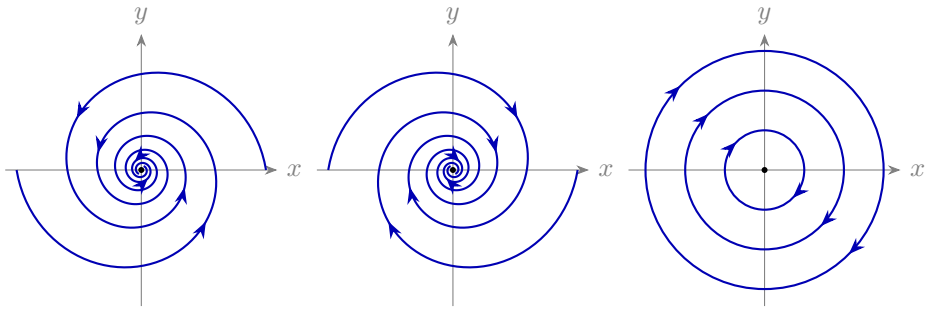
[19] https://www.youtube.com/watch?v=0EcFQ7j4qo4 (This was the last one I promise.)

[20] For some reason it's the only possible one because a complex lambda would make for a dim 4 situation?. If λ is an e-value of $E \in \mathbb{R}^{2\times 2}$, then $\bar{\lambda}$ is also one. That is not possible (i guess?) because it's four dimensions. Così ha detto a lezione, io non ho capito.



$a = 0.25,\ b = 1.5$ $a = 0.25,\ b = -1.5$ $a = 0,\ b = 1.5$

(a) Unstable spiral, $a > 0$, $b > 0$ (counterclockwise). (b) Unstable spiral, $a > 0$, $b < 0$ (clockwise). (c) Center, $a = 0$, $b > 0$ (counterclockwise).



$a = -0.25,\ b = 1.5$ $a = -0.25,\ b = -1.5$ $a = 0,\ b = -1.5$

(d) Stable spiral, $a < 0$, $b > 0$ (counterclockwise). (e) Stable spiral, $a < 0$, $b < 0$ (clockwise). (f) Center, $a = 0$, $b < 0$ (clockwise).

Figure 1.15: Phase portraits of $\dot{z} = Bz$ with $e^{tB} = e^{at}R_{bt}$. a determines whether trajectories move away from ($a > 0$) or towards ($a < 0$) the origin, while b determines the sense of rotation. Arrows indicate the direction of time.

is the following:

$$\begin{bmatrix} e^{\lambda t} & te^{\lambda t} \\ 0 & e^{\lambda t} \end{bmatrix} \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} = \begin{bmatrix} x_0 e^{\lambda t} + y_0 te^{\lambda t} \\ y_0 e^{\lambda t} \end{bmatrix}$$

which means that:

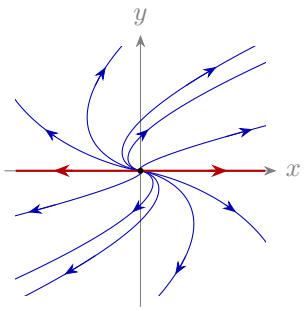
$$\begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = \begin{bmatrix} x_0 e^{\lambda t} + y_0 te^{\lambda t} \\ y_0 e^{\lambda t} \end{bmatrix}$$

We follow the standard protocol of eliminating the time dependencies for plotting reasons, by solving for time and substituting. Quite clearly, the easy one to solve for time is y here. Once done, you should find:

$$t = \frac{log(\frac{y}{y_0})}{\lambda}$$

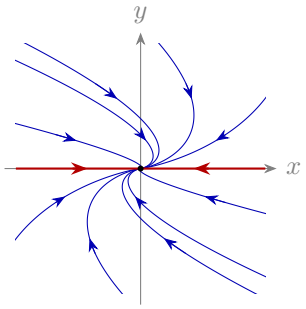
Once plugged into the equation, x(y) is the following, as plotted in Figure 1.16:

$$x = \frac{x_0}{y_0} y + \frac{1}{\lambda} y \log(\frac{y}{y_0})$$



$\lambda = 1$

(a) Unstable degenerate node, $\lambda > 0$.



$\lambda = -1$

(b) Stable degenerate node, $\lambda < 0$.

Figure 1.16: Phase portraits of $\dot{x} = \lambda x + y$, $\dot{y} = \lambda y$. Blue curves are the trajectories with $y_0 \neq 0$; red half-lines are the trajectories on the x-axis ($y_0 = 0$), the only invariant line. Arrows indicate the direction of increasing time.

1.4 Linearized ODEs around a fixed point

Suppose one has the following IVP:

$$\begin{cases} \dot{x} = f(x), & x \in M \subseteq \mathbb{R}^d \\ x(0) = x_0 \end{cases}$$

where $\bar{x}$ is a fixed point. Note that:

$$\bar{x} \text{ is a fixed point} \Leftrightarrow f(\bar{x}) = 0 \Leftrightarrow (\phi^t(\bar{x}) = \bar{x} \forall t \Leftrightarrow x(t, \bar{x}) \text{ is a solution})^{21}$$

We assume that $f \in C^1$. This implies that if x is close to $\bar{x}$, we can use the differential and also the jacobian matrix to approximate as follows:

$$x \text{ and } \bar{x} \text{ are close} \Rightarrow f(x) = f(\bar{x}) + Df_{\bar{x}}(x - \bar{x}) + o(|x - \bar{x}|) \approx 0 + Jf_{\bar{x}} \cdot (x - \bar{x})^{22}$$

[21] The last one meaning that if you start from $\bar{x}$ you stay in $\bar{x}$.

[22] (as a column vector)

where the Jacobian is defined as:

$$J_f(x) = \frac{\partial f}{\partial x} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \frac{\partial f_m}{\partial x_2} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}$$

By doing so, we may linearize the ODE as follows:

$$\dot{x} = J_f(x - \bar{x}) \Rightarrow \dot{y} = J_f y, \qquad y = x - \bar{x}$$

We have to remember that we are dealing with an approximated solution, extremely close to $\bar{x}$. We will later see that, in some cases, the linear system and the non-linear system are conjugate to each other for any time, if we keep close to the fixed point. To do that, we must first introduce hyperbolic points.

Definition 1.10 The fixed point $\bar{x}$ is called hyperbolic if $D_{\bar{x}}f$ has no eigenvalue on the imaginary line, meaning that, letting λ be an eigenvalue, $Im\lambda = 0$ if $Re\lambda = 0$.

In 2D, hyperbolic behaviour is the one we found in Example 1.9: one of the directions is contracting, while the other one is expanding. To be precise, situations in which the directions are both contracting (or expanding?) but with different magnitudes, as we found in the same example, are still considered hyperbolic. Because of this, we define truly hyperbolic points.[23]

Definition 1.11 The fixed point $\bar{x}$ is called truly hyperbolic if $D_{\bar{x}}f$ has one eigenvalue with a positive real part and one with a negative real part.

To properly understand what is NOT hyperbolic, you may look at Example 1.10, which gave us a spiral. As a homework, consider $A = [i0i0] \Rightarrow B = [0 - 110]$ and verify that e^{tB} is not hyperbolic.

We are now ready to explain how legitimate the linearization we did at the start of the section is.

Theorem 1.2 (Hartman-Grobmann theorem for flows) *Let ϕ^t be the "partial flow"[24] of $\dot{x} = f(x)$ where*

$$f(x) : \Omega \to \mathbb{R}^d,\ \Omega \subseteq \mathbb{R}^d,\ f(x) \in C^1$$

Let $\bar{x}$ be a hyperbolic fixed point and let

$$\tilde{\phi}^t(y) = e^{tA}y, \qquad A = Jf_{\bar{x}}$$

[23] The professor made this distinction, but it is very uncommon in the literature.
[24] Meaning not defined for every t

be the flow of $\dot{y} = Ay$.

Then, there are a neighbourhood U *of* $\bar{x}$ *and a neighbourhood* V *of 0 and a homeomorphism*[25] h *from* U *to* V *such that* $\phi^t(x) = (h^{-1} \circ \tilde{\phi}^t \circ h)(x)$ *for all* $x \in U$ *and a* $t \in \mathbb{R}$ *such that* $\phi^t(x) \in U$*, which in turn implies (it's actually an iff*[26]*) that* $h(x)$ *is in* U *and* $\tilde{\phi}^t(h(x))$ *is in* V.

In mathematical terms:

$$\phi^t \text{ is the partial flow of } \dot{x} = f(x),\ f(x) : \Omega \to \mathbb{R}^d,\ \Omega \subseteq \mathbb{R}^d,\ f(x) \in C^1,$$
$$\bar{x} \text{ is a hyberbolic fixed point, } \tilde{\phi}^t(y) = e^{tA}y,\ A = Jf_{\bar{x}} \text{ is the flow of } \dot{y} = Ay \Rightarrow$$
$$\exists U \text{ nbhd of } \bar{x},\ V \text{ nbhd of } 0,\ h : U \to V \text{ homeomorfism } |$$
$$\phi^t(x) = (h^{-1} \circ \tilde{\phi}^t \circ h)(x) \forall x \in U \wedge \exists t \in \mathbb{R} | \phi^t \in U (\Leftrightarrow h(x) \in V, \tilde{\phi}^t(h(x)) \in V)$$

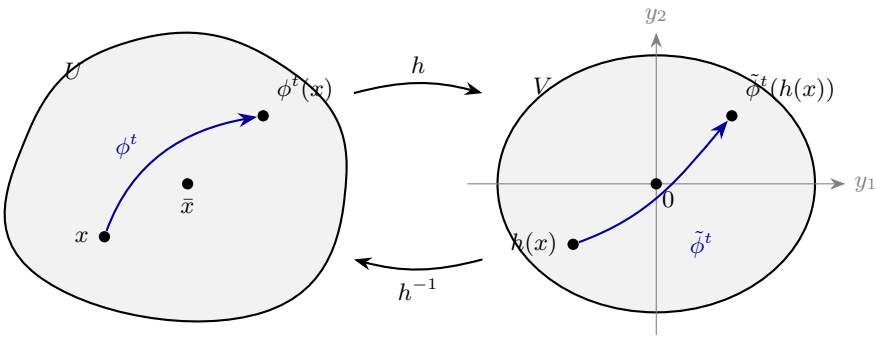


Figure 1.17: The homeomorphism $h\colon U \to V$ maps the trajectory of the nonlinear flow ϕ^t in the neighbourhood U of the hyperbolic fixed point $\bar{x}$, going from x to $\phi^t(x)$, onto the trajectory of the linear flow $\tilde{\phi}^t(y) = e^{tA}y$ in the neighbourhood V of the origin, going from $h(x)$ to $\tilde{\phi}^t(h(x))$. Hence $\phi^t = h^{-1} \circ \tilde{\phi}^t \circ h$ as long as the trajectory stays in U.

Basically, the theorem tells us that solutions are mapped in the linearized problem, when we limit ourselves to U and V. It's important thet the point is hyperbolic. Say it's not hyperbolic, say it's actually attracting: that works at all times, ??

Observation 1.6 Although f is differentiable, h is not guaranteed to be a diffeomorphism[27], which means that you don't know the shape of the phase diagram in U just because you know it in V as trivially: if it was differentiable, you would have a simpler transformation.

[25] A continuous bijective function with continuous inverse

[26] If and only if

[27] A differentiable bijective function with differentiable inverse.

Application: Lotka-Volterra equation

Now, we shall see an application of the theory, the Lotka-Volterra model for animal populations. The model is the following:

$$\begin{cases} \dot{x} = x(\alpha - \beta y) \\ \dot{y} = -y(\gamma - \delta x) \qquad \alpha, \beta, \gamma, \delta > 0 \\ x(0) = x_0 \\ y(0) = y_0 \qquad x,\, y \in \Omega,\, \Omega \subseteq (\mathbb{R}^+)^2 \end{cases}$$

where x is a population of preys while y is a population of predators.

Let's look for the fixed points. The vector field is given by $f(x, y) = (\alpha x - \beta xy, -\gamma y + \delta xy)$, so we may proceed as follows:

$$\begin{cases} x(\alpha - \beta y) = 0 \\ -y(\gamma - \delta x) = 0 \end{cases} \Rightarrow (x_0, y_0) = (0, 0),\, (x_1, y_1) = \left(\frac{\gamma}{\delta}, \frac{\alpha}{\beta}\right)$$

Let's now linearize around the fixed points. We shall start by finding the Jacobian:

$$Df_{x,y} = Jf_{x,y} = \begin{bmatrix} \partial_x f_x & \partial_y f_x \\ \partial_x f_y & \partial_y f_y \end{bmatrix} = \begin{bmatrix} \alpha - \beta y & -\beta x \\ \delta y & -\gamma + \delta x \end{bmatrix}$$

And then we compute it at the fixed points:

$$Jf_{x_0,y_0} = \begin{bmatrix} \alpha & 0 \\ 0 & -\gamma \end{bmatrix}$$

$$Jf_{x_1,y_1} = \begin{bmatrix} 0 & -\beta\frac{\gamma}{\delta} \\ \delta\frac{\alpha}{\beta} & 0 \end{bmatrix} = A$$

We may ask ourselves if this is a hyperbolic point. That can be shown by computing the eigenvalues:

$$\det(A - \lambda I) = \lambda^2 + \alpha\gamma = 0 \Rightarrow \lambda = \pm i\sqrt{\alpha\gamma}$$

So, x_1, y_1 is not hyperbolic.

Proposition 1.8 *This system has a constant of motion Q:*

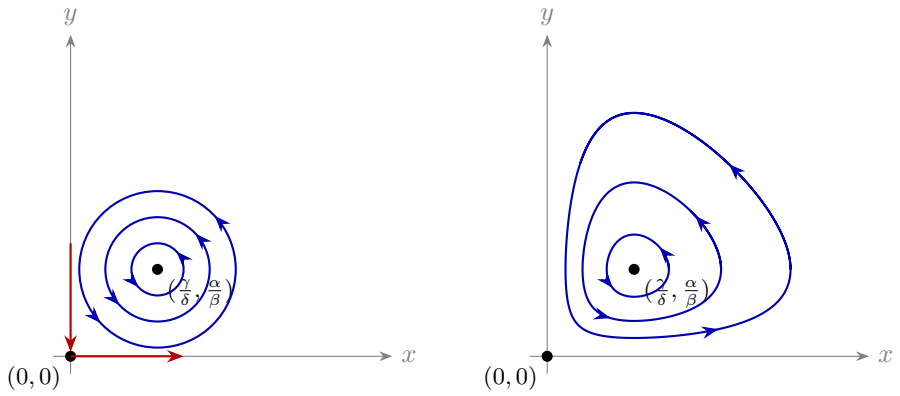
$$Q : (\mathbb{R}^+)^2 \to \mathbb{R}, \qquad Q = -\delta x + \gamma log x - \beta y + \alpha log y$$

Proof. We aim at proving that for every solution $x(t)$, $y(t)$, $\frac{dQ}{dt} = 0$.

So, by substituting the values of $\dot{x}$ and $\dot{y}$, we may easily see that:

$$\dot{Q} = -\delta\dot{x} + \gamma\frac{\dot{x}}{x} - \beta\dot{y} + \alpha\frac{\dot{y}}{y} = -\alpha\delta x + \beta\delta xy + \alpha\gamma - \beta\gamma y + \beta\gamma y - \beta\delta xy - \alpha\gamma + \alpha\delta x = 0$$

□



(a) Linearized system: a saddle at $(0,0)$ and a centre at $(\frac{\gamma}{\delta}, \frac{\alpha}{\beta})$ (closed orbits).

(b) Nonlinear system: the orbits are closed curves around $(\frac{\gamma}{\delta}, \frac{\alpha}{\beta})$ and never reach $(0,0)$.

Figure 1.18: Phase portraits of the Lotka-Volterra model with $\alpha = \beta = \gamma = \delta = 1$. Arrows indicate the direction of increasing time (counterclockwise around the second fixed point).

1.5 Linear maps

Let $T(x)$ be a linear map defined as follows[28]:

$$T(x) = Bx,\ x \in \mathbb{R}^d,\ B \in \mathbb{R}^{d \times d} \tag{1.10}$$

The system solves itself. The orbit O is the following[29]:

$$O(x) = (B^n x)_{n \in \mathbb{N}}$$

The set of fixed points of this map is given by:

$$Fix(T) = Ker(B - I)$$

which directly implies that $0 \in Fix(T)$. In other words, finding the fixed points means to solve the following equation:

$$Bx = x \Rightarrow (B - I)x = 0$$

1.5.1 Exhaustive list of examples in 2D

Example 1.12 Let the B in Equation 1.10 be defined as follows:

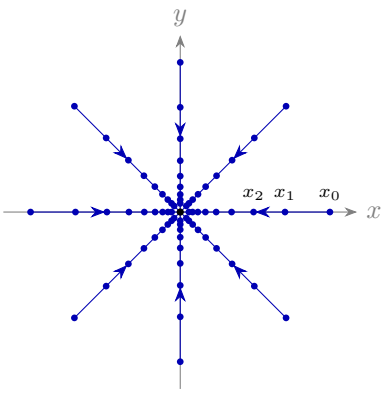
$$B = \begin{bmatrix} \alpha & 0 \\ 0 & \alpha \end{bmatrix} = \alpha I$$

[28] Assume B to be in Jordan form.

[29] $O(x)$ may be occasionally called $x_n = B^n x_0$ from now on.

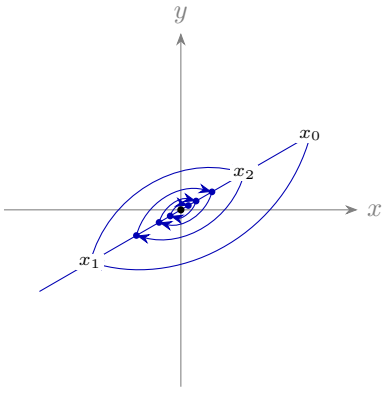
We may easily find that:

$$\begin{bmatrix} x_n \\ y_n \end{bmatrix} = \begin{bmatrix} \alpha^n x_0 \\ \alpha^n y_0 \end{bmatrix}$$



$\alpha = 0.7$

(a) $0 < \alpha < 1$: the points lie on straight lines and approach the origin monotonically.



$\alpha = -0.7$

(b) $-1 < \alpha < 0$: the points alternate between the two sides of the origin on the same line.

Figure 1.19: Orbits of $x_{n+1} = \alpha x_n$ for $B = \alpha I$. Dots are the discrete points $x_n = \alpha^n x_0$. Since $|\alpha| < 1$, the origin is attracting and the distance to it decreases exponentially, as $|\alpha|^n$.

Example 1.13 Let the B in Equation 1.10 be defined as follows:

$$B = \begin{bmatrix} \alpha & 0 \\ 0 & \beta \end{bmatrix}, \qquad \alpha, \beta \in \mathbb{R}$$

We may, again, easily find the solution to be the following:

$$\begin{bmatrix} x_n \\ y_n \end{bmatrix} = \begin{bmatrix} \alpha^n x_0 \\ \beta^n y_0 \end{bmatrix}$$

To find the line, we want to eliminate n from the system. We may easily find:

$$\frac{\log\left(\left|\frac{x_n}{x_0}\right|\right)}{\log(|\alpha|)} = n = \frac{\log\left(\left|\frac{y_n}{y_0}\right|\right)}{\log(|\beta|)}$$

By equating them:

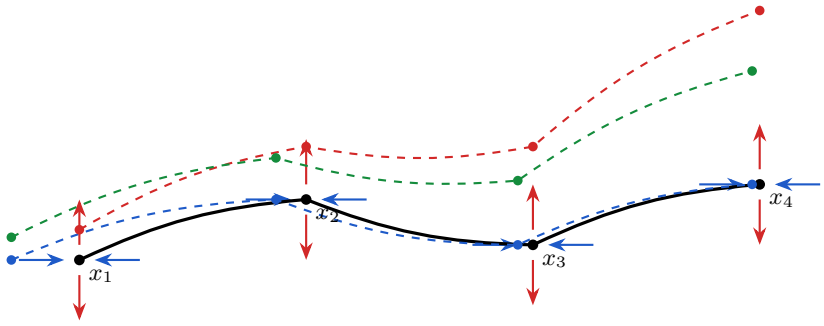
$$\log\left(\left|\frac{y_n}{y_0}\right|\right) = \frac{\log|\beta|}{\log|\alpha|}\log\left(\left|\frac{x_n}{x_0}\right|\right) \Rightarrow \left|\frac{y_n}{y_0}\right| = \left|\frac{x_n}{x_0}\right|^{\frac{\log|\beta|}{\log|\alpha|}} \Rightarrow$$

$$|y| = C|x|^{\frac{\log|\beta|}{\log|\alpha|}}, \qquad C = \left|\frac{y_0}{x_0}\right|$$

This example may also serve to introduce the concept of invariant curves.

Definition 1.12 Any curve which contains an entire orbit is called an invariant curve.

Also, this whole hyperbolic behaviour can explain why we observe chaos: as much as that may be pub talk, the idea is easily inferrable from what we know. Imagine you are trying to repeat a trajectory: you mill, almost surely, miss the exact initial conditions. This will be mitigated on contracting directions, in which the trajectory will get closer. But say one direction is expanding: then the new trajectory, in that direction, will exponentially get further away from the points of the initial trajectory, thus making every one of them wildly different over some minor difference in initial conditions, as shown in the picture below. Obviously, in the real world, those differences may be minimal but will nonetheless be unpredictable. In general, truly hyperbolic points are important in physics because expanding directions are really common in Hamiltonian dynamics.



Example 1.14 Let the B in Equation 1.10 be defined as follows:

$$B = \begin{bmatrix} \alpha & 1 \\ 0 & \alpha \end{bmatrix}, \qquad \alpha \in \mathbb{R}$$

$$B^n = \begin{bmatrix} \alpha^n & n\alpha^{n-1} \\ 0 & \alpha^n \end{bmatrix} \Rightarrow \begin{bmatrix} x_n \\ y_n \end{bmatrix} = \begin{bmatrix} x_0\alpha^n + y_0 n\alpha^{n-1} \\ y_0\alpha^n \end{bmatrix}$$

Assuming $\alpha \in \mathbb{R}^+/\{1\}$[30]. We use the usual technique to find the invariant curve:

$$n = \frac{\log\left(\frac{y_n}{y_0}\right)}{\log(\alpha)}$$

Plug that in x_n and you get:

$$x_n = \frac{y_n}{y_0}x_0 + \frac{\log\left(\frac{y_n}{y_0}\right)}{\log(\alpha)}\frac{1}{\alpha}\frac{y_n}{y_0} \Rightarrow x = \frac{x_0}{y_0}y + \frac{y\log\left(\frac{y_n}{y_0}\right)}{\alpha\log(\alpha)}$$

[30] Homework: do the case $\alpha \in \mathbb{R}^-/-1$.

(disegno dell'aspetto per alpha maggiore di 1 e anche minore ma maggiore di 0, stessa cosa che per le ODE comunque)

In general, the equation is given by:

$$x = y(C + \frac{\log|y|}{\alpha \log|\alpha|})$$

Example 1.15 Let the B in Equation 1.10 be defined as follows:

$$B = \begin{bmatrix} c+id & 0 \\ 0 & c-id \end{bmatrix} = \begin{bmatrix} Re^{i\theta} & 0 \\ 0 & Re^{-i\theta} \end{bmatrix}, \; R = \sqrt{c^2+d^2}, \; \theta = \arctan\left(\frac{d}{c}\right)$$

Taking the usual $D = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & i \\ i & 1 \end{bmatrix}$, we do:

$$D^{-1}BD = R \begin{bmatrix} cos(\theta) & -sin(\theta) \\ sin(\theta) & cos(\theta) \end{bmatrix} = \hat{B}$$

We change coordinates, $z = D^{-1}x$, which impiles

$$x_{n+1} = Bx_n \iff z_{n+1} = \hat{B}z_n$$

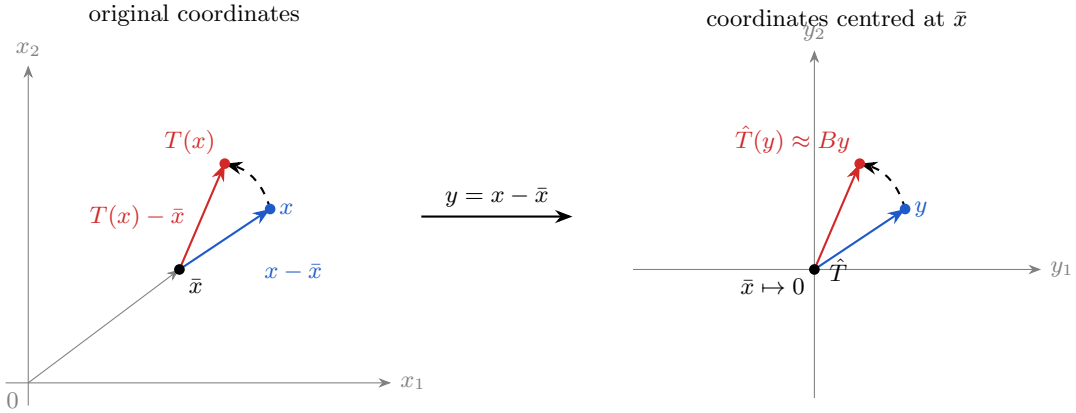
Witht the usual abuse of notation:

$$\begin{bmatrix} x_n \\ y_n \end{bmatrix} = \hat{B} \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} = R^n R_{n\theta} \frac{x_0}{y_0}$$

[31]

The reason we studied these examples is to approximate other maps.

1.5.2 Linearization of a map around a fixed point



[31] HW find the equation of the invariant curve (e.g. in polar coordinates)

Suppose you have a map $T : M \to M$, $M \subseteq \mathbb{R}^d$ and that T is C^1, and suppose you have a fixed point $\bar{x} \in Fix(T)$.

For x close to $\bar{x}$, $T(x) = T(\bar{x}) + J_{\bar{x}}T(x - \bar{x}) + o(|x - \bar{x}|)$, calling the Jacobian B we can say:

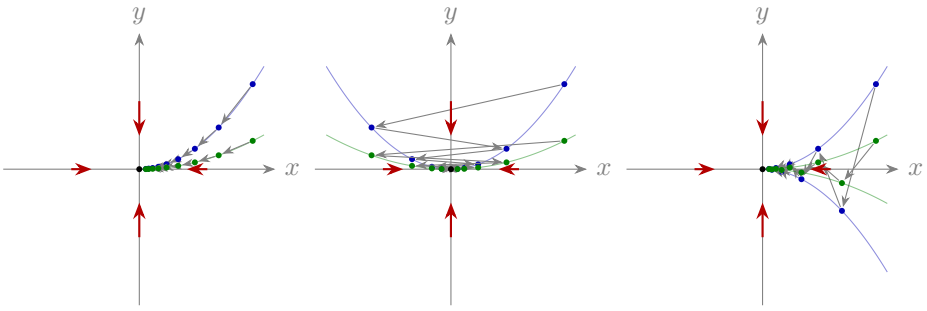
$$T(x) - \bar{x} \approx B(x - \bar{x})$$

If we change coordinates, calling $y = x - \bar{x}$, then

$$\hat{T}(y) = T(y + \bar{x}) - \bar{x}$$

and we can say

$$\hat{T}(y) \approx By$$

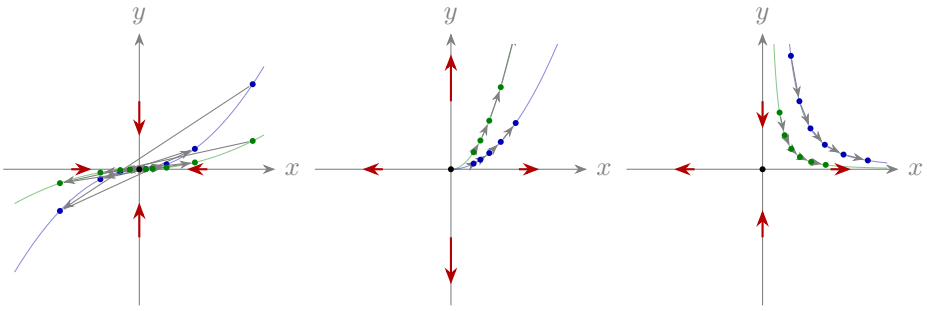


$\alpha = 0.7,\ \beta = 0.49$ $\alpha = -0.7,\ \beta = 0.49$ $\alpha = 0.7,\ \beta = -0.49$

(a) Attractive node, $0 < \beta < \alpha < 1$.

(b) $-1 < \alpha < 0 < \beta$: points jump around the y-axis.

(c) $\alpha > 0 > \beta$: points jump around the x-axis.



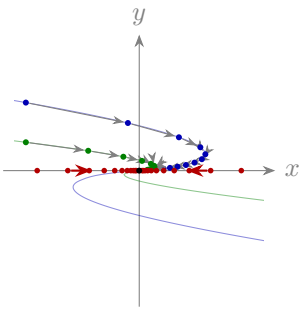
$\alpha = -0.7,\ \beta = -0.49$ $\alpha = 1.3,\ \beta = 1.69$ $\alpha = 1.3,\ \beta = 0.6$

(d) $\alpha, \beta < 0$: points jump around the origin (quadrants I, III, I, III, ...).

(e) Repulsive node, $1 < \alpha < \beta$.

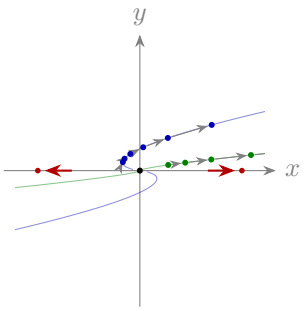
(f) Saddle, $\beta < 1 < \alpha$: the y-axis contracts, the x-axis expands.

Figure 1.20: Orbits of $(x_{n+1}, y_{n+1}) = (\alpha x_n, \beta y_n)$. Dots are the points $(\alpha^n x_0, \beta^n y_0)$ for two initial conditions, grey arrows are the jumps $x_n \to x_{n+1}$, and the curves are $|y| = C|x|^{\log|\beta|/\log|\alpha|}$. Red arrows on the axes show the effect of one step: inwards for contraction, outwards for expansion.



$\alpha = 0.7$

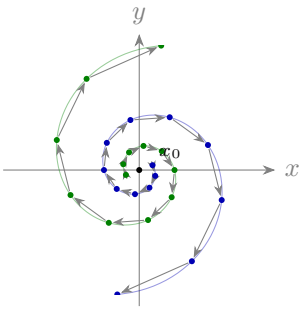
(a) $0 < \alpha < 1$: attractive, the orbits go to the origin.



$\alpha = 1.4$

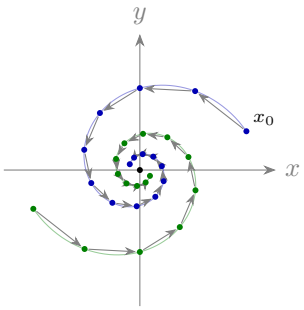
(b) $\alpha > 1$: repulsive, the orbits leave the origin.

Figure 1.21: Orbits of $(x_{n+1}, y_{n+1}) = (\alpha x_n + y_n,\ \alpha y_n)$. Dots are the points P_n for two initial conditions (blue and green), thin grey arrows are the jumps $P_n \to P_{n+1}$, and the faded curves are the invariant curves $x = \frac{x_0}{y_0} y + \frac{y \log(y/y_0)}{\alpha \log \alpha}$. The red dots and arrows on the x-axis show the only invariant line.



$R = 1.2,\ \theta = -40°$

(a) $R > 1,\ \theta < 0$: repulsive, the points spiral outwards clockwise.



$R = 0.85,\ \theta = 35°$

(b) $0 < R < 1,\ \theta > 0$: attractive, the points spiral inwards counterclockwise.

Figure 1.22: Orbits of $z_{n+1} = \hat{B} z_n$ with $\hat{B} = R\, R_\theta$. Dots are the points $z_n = R^n R_{n\theta} z_0$ for two initial conditions (blue and green, one the 180° rotation of the other), thin grey arrows are the jumps $z_n \to z_{n+1}$, and the faded curves are the logarithmic spirals interpolating them.

Chapter 2

Topological dynamics

Chapter 3

Ergodic theory

Chapter 4

Possible developments

Appendix

Matlab code for the logistic map

I took the liberty to use a slightly different code from the professor for that simulation:

Listing 4.1: Logistic Map Script

```
clear all

close all

set(groot, 'defaultFigureColor', 'w', ...
    'defaultAxesColor', 'w', ...
    'defaultAxesXColor', 'k', ...
    'defaultAxesYColor', 'k', ...
    'defaultTextColor', 'k');

N=200; % number of iterations

r=2.6;

x0=0.95;


xm=linspace(0,1,100);

ym=map(r,xm);

x = zeros(1,N);

xp=zeros(1,2*N+1);

yp=zeros(1,2*N+1);

x(1)=x0; % initial condition (can be anything from 0
```

```
    to 1)
29
30  xp(1)=x0;
31
32
33  for n = 1:N
34
35      x(n+1)=map(r,x(n));
36
37      xp(2*n)=xp(2*n-1);
38
39      yp(2*n)=x(n+1);
40
41      xp(2*n+1)=x(n+1);
42
43      yp(2*n+1)=x(n+1);
44
45  end
46
47  % Define the double-time map function
48  map2 = @(r, s) map(r, map(r, s));
49  ym2 = map2(r, xm);
50
51  % Compute trajectory and cobweb points for the double-
        time map
52  x2 = zeros(1, N);
53  xp2 = zeros(1, 2*N+1);
54  yp2 = zeros(1, 2*N+1);
55  x2(1) = x0;
56  xp2(1) = x0;
57
58  for n = 1:N
59      x2(n+1) = map2(r, x2(n));
60      xp2(2*n) = xp2(2*n-1);
61      yp2(2*n) = x2(n+1);
62      xp2(2*n+1) = x2(n+1);
63      yp2(2*n+1) = x2(n+1);
64  end
65
66  % Create a single figure window with subplots
67  figure;
68
69  % First panel: Map function and Cobweb diagram
70  subplot(1, 2, 1);
71  plot(xm, xm, 'k--', 'LineWidth', 1);     % Identity
        line y = x
```

```
hold on;
plot(xm, ym, 'r', 'LineWidth', 1.5);         % Single-time map curve
plot(xp, yp, 'b');                            % Single map cobweb trajectory
title('T(x)');
xlabel('x_n');
ylabel('x_{n+1}');
legend('y = x', 'Single Map', 'Cobweb 1x');
grid on;

%Second panel: Double map
%subplot(1, 3, 2);
%plot(xm, xm, 'k--', 'LineWidth', 1);      % Identity line y = x
%hold on;
%plot(xm, ym2, 'm', 'LineWidth', 1.5);       % Double-time map curve
%plot(xp2, yp2, 'g');                        % Double map cobweb trajectory
%title('T^2(x)');
%xlabel('x_n');
%ylabel('x_{n+1}');
%legend('y = x', 'Double Map', 'Cobweb 2x');
%grid on;

% Second/third panel: Time series trajectory x(n)
subplot(1, 2, 2);
plot(x, '.-');
xlim([0 200]);
title('Trajectory x(n)');
xlabel('n');
ylabel('x_n');
grid on;

function m=map(r,s)
m=r*s.*(1-s);
%    m0=r*s.*(1-s);   % this is the double-time map
%    m=r*m0.*(1-m0);
end
```

One quick note: to display also the squared map, one must uncomment the middle panel and then change all the subplot() have 3 as the middle number, and also as the last one for the trajectory panel.